Bunkbot study 4 (#282517)

Author(s)
This pre-registration is currently anonymous to enable blind peer-review.
It has 4 authors.

Pre-registered on:
2026/03/31 08:52 (PT)

1) Have any data been collected for this study already?
It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?
This study asks two main questions.

1. Does the asymmetry between AI bunking and debunking vary across models?
The primary confirmatory outcome for this question is belief change coded in the direction of intended persuasion, so higher values always mean movement in the assigned direction.

2. Do these conversational effects extend to what people would put into the information ecosystem?
We test whether the intervention changes the direction and shareability of participants' social media posts.

We predict that:

1. Models will differ in the relative effectiveness of bunking versus debunking on belief change.
2. Social-post outcomes will also move in the assigned direction of persuasion.
3. Models that are more truth-constrained / accurate will show more debunk-favoring asymmetry than less accurate models.

3) Describe the key dependent variable(s) specifying how they will be measured.
Aligned belief change post belief - pre belief, multiplied by +1 in bunk and -1 in debunk.

Aligned change from old post to new post in weighted sharing (stance of new post * sharing of new post) - (stance of old post * sharing of old post), coded so higher values always indicate movement in the assigned direction.

Aligned change in stance without weighting by sharing new post stance - old post stance, coded so higher values always indicate movement in the assigned direction.

Conversation accuracy / veracity Average veracity of all assistant factual claims in the conversation.
Aligned change in weighted sharing of the original post The original post is LLM-scored for stance toward the focal conspiracy on a 0-100 scale, recentered to -50 to +50, multiplied by sharing likelihood, and then coded so higher values always indicate movement in the assigned direction.

4) How many and which conditions will participants be assigned to?
Participants will be randomly assigned with equal probability to one of 8 cells in a 2 x 4 between-subjects design:

Persuasion direction
Bunk
Debunk
Model type
google/gemini-3-pro-preview
openai/gpt-5.2
anthropic/claude-opus-4-6
x-ai/grok-4

All participants will identify a conspiracy theory they are uncertain about, write an initial social media post, report how likely they would be to share it, converse with the assigned AI, then report post-treatment belief and social-post outcomes.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.
Estimate aligned_belief_change ~ persuasion_direction * model_type using OLS with robust standard errors.

Estimate aligned_original_post_share_change ~ persuasion_direction * model_type
Estimate aligned_new_minus_old_weighted ~ persuasion_direction * model_type
Estimate aligned_stance_change ~ persuasion_direction * model_type

For the raw, non-direction-coded versions of the social-post outcomes, report overall pre-post change, change by persuasion direction, and change by model.
If these ecosystem outcomes show meaningful model interactions, report the condition-by-model breakdowns.

If model heterogeneity in aligned belief change is observed, relate belief-change estimates to conversation accuracy.

In these ecosystem analyses, the goal is not mainly to estimate the bunk-versus-debunk difference coefficient, but to show whether either treatment arm substantially changes what participants would contribute to the information ecosystem.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.
We will exclude:

Participants who fail attention checks.
Participants whose focal-conspiracy response is invalid or does not indicate genuine uncertainty.
Participants whose pre-treatment focal-belief rating is outside the intended uncertainty range after the survey reroute logic.
Participants with missing or unusable treatment data.
Duplicate or otherwise invalid respondent IDs.
Participants who fail bot checks.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.
We will recruit N = 2400 valid participants, randomly assigned equally across the 8 cells (n = 300 per cell). If exclusions occur during fielding, recruitment will continue until 2400 valid completes are obtained.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)
Due to an oversight, we wrote and finalized the pre-registration before data collection, but did not officially submit it to aspredicted until data was actively being collected.